

WEEK 4 – POWER BI EDA

SHOPIFY STOCK DATA ANALYSIS

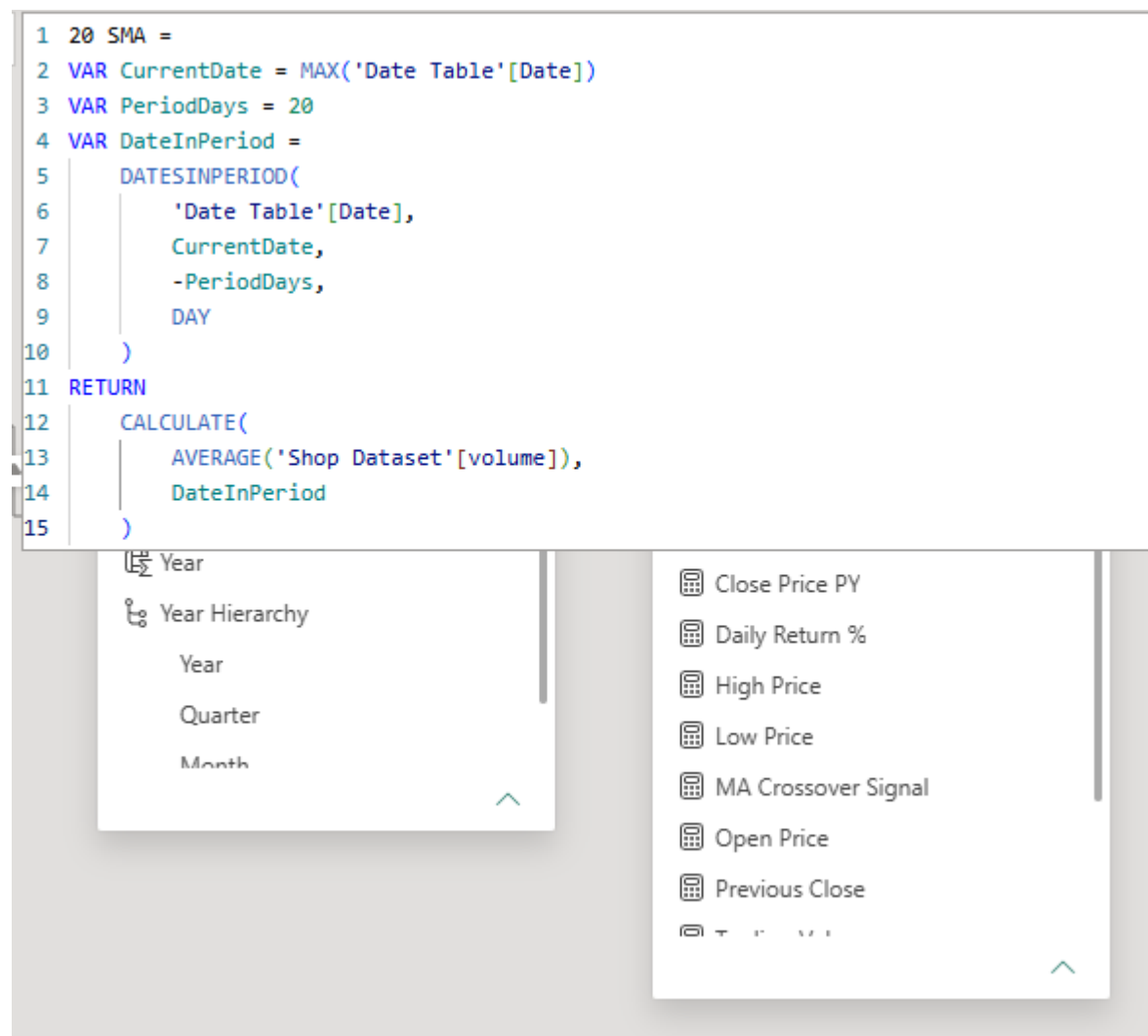
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Project: Shopify Stock Analysis using Power BI

1. Simple Moving Average (SMA) – 20 Days and 50 Days

Two Simple Moving Average (SMA) measures were created using DAX to calculate the 20-day and 50-day moving averages of trading volume. These measures help analyze the average trading volume over the selected periods.

```
1 20 SMA =
2  VAR CurrentDate = MAX('Date Table'[Date])
3  VAR PeriodDays = 20
4  VAR DateInPeriod =
5      DATESINPERIOD(
6          'Date Table'[Date],
7          CurrentDate,
8          -PeriodDays,
9          DAY
10     )
11 RETURN
12     CALCULATE(
13         AVERAGE('Shop Dataset'[volume]),
14         DateInPeriod
15     )
```

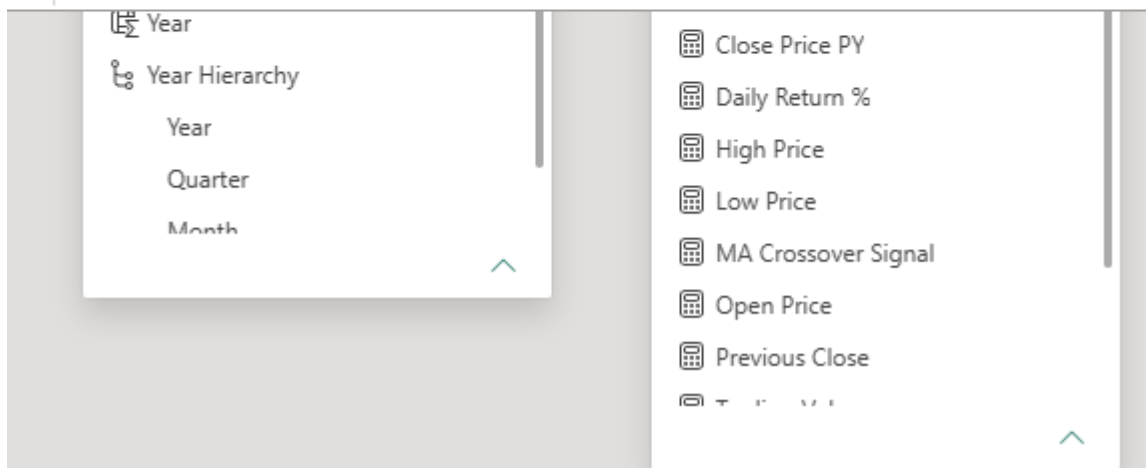


The screenshot displays the Power BI Desktop interface. The top section shows the DAX formula for a measure named '20 SMA'. The formula uses the DATESINPERIOD function to define a date range ending at the current date and extending back 20 days, then uses CALCULATE to average the trading volume from the 'Shop Dataset' over this period. Below the formula bar, the Fields pane is open, showing two categories of measures. The first category, 'Date Table', includes 'Year', 'Year Hierarchy', 'Year', 'Quarter', and 'Month'. The second category, 'Shop Dataset', includes 'Close Price PY', 'Daily Return %', 'High Price', 'Low Price', 'MA Crossover Signal', 'Open Price', and 'Previous Close'.

```

1 50 SMA =
2 VAR CurrentDate = MAX('Date Table'[Date])
3 VAR PeriodDays = 50
4 VAR DateInPeriod =
5     DATESINPERIOD(
6         'Date Table'[Date],
7         CurrentDate,
8         -PeriodDays,
9         DAY
10    )
11 RETURN
12     CALCULATE(
13         AVERAGE('Shop Dataset'[volume]),
14         DateInPeriod
15    )

```



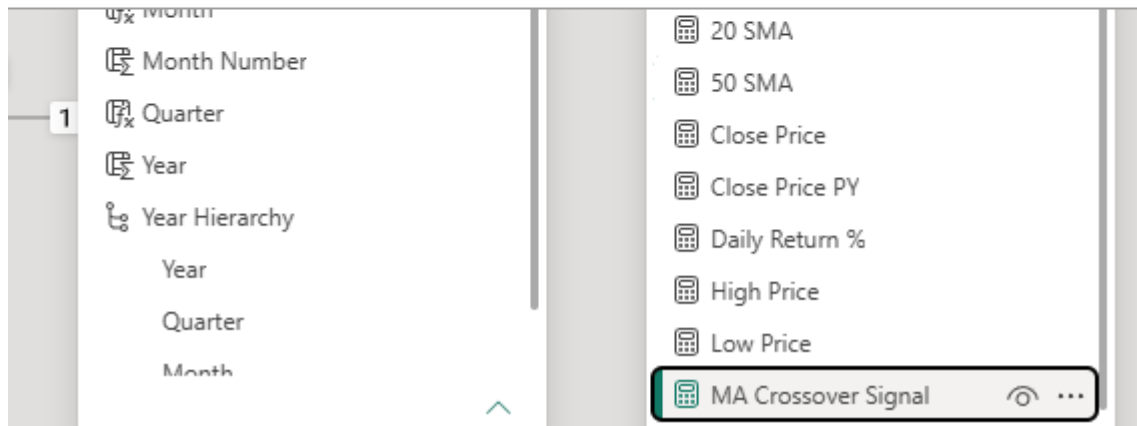
2. Moving Average Crossover Signal

The Moving Average Crossover Signal was created using DAX by comparing the 20-day SMA with the 50-day SMA. The result identifies whether the 20 SMA is above or below the 50 SMA.

```

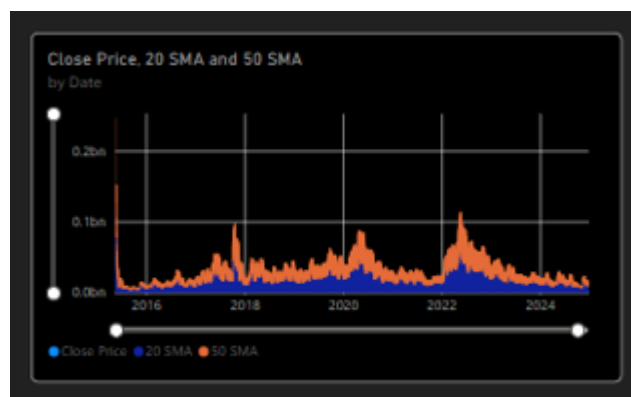
1 MA Crossover Signal =
2 IF(
3   ISBLANK([20 SMA]) ||
4   ISBLANK([50 SMA]),
5   BLANK(),
6   IF(
7     [20 SMA] > [50 SMA],
8     "Bullish (20 > 50)",
9     "Bearish (20 < 50)"
10  )
11 )

```



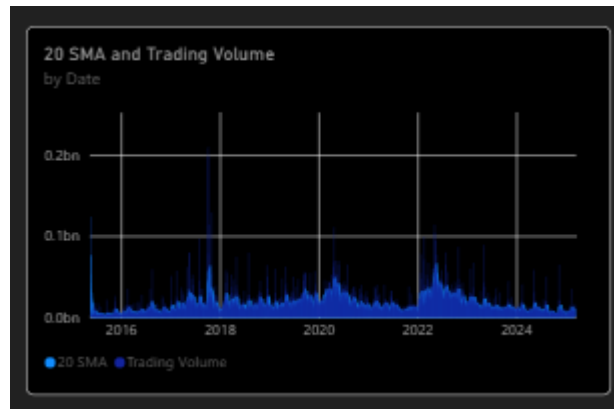
3. Interactive OHLC & Price Trend Chart

An interactive price trend chart was created using Date, Close Price, 20 SMA, and 50 SMA. The Date axis was configured as Continuous, and a Zoom Slider was enabled to allow users to examine different time periods.



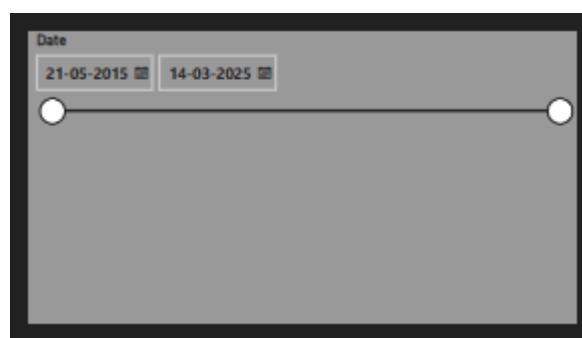
4. Trading Volume & Volatility Visual

A clustered column chart was created to analyze Trading Volume and the 20-day SMA over time. The Date axis was configured as Continuous to maintain consistent time-series analysis.



5. Date Range Slicer

A Date Range Slicer was added using the Date field and configured in Between mode. This allows a specific start date and end date to be selected for the analysis.



6. Relative Date Slicer

A Relative Date Slicer was added to provide dynamic filtering based on a selected relative time period.

Date

Last 1 Select

No filters applied

7. Price Metric Parameter

A Field Parameter named Price Metric was created to allow dynamic selection of different stock price measures. The parameter includes Close Price, High Price, Low Price, and Open Price.

Price Metric

☐ Close Price

☐ High Price

☐ Low Price

☐ Open Price

8. Custom Tooltip – Stock_Tooltip

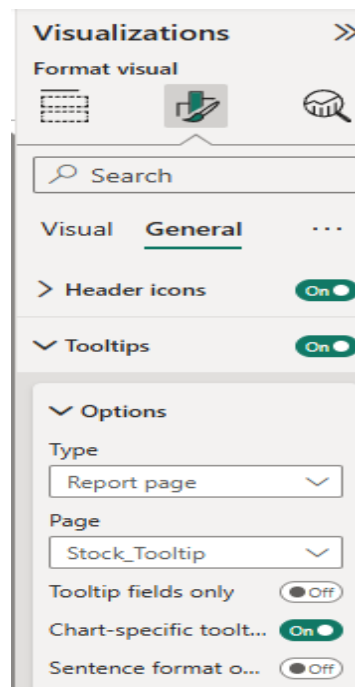
A custom tooltip page named Stock_Tooltip was created to display detailed stock information when the user hovers over the price trend chart. The tooltip contains Open Price, High Price, Low Price, Close Price, Trading Volume, and Daily Return %.

Open Price	High Price	Low Price	Close Price	Trading Volume
48.46	176.29	1.85	48.46	409125

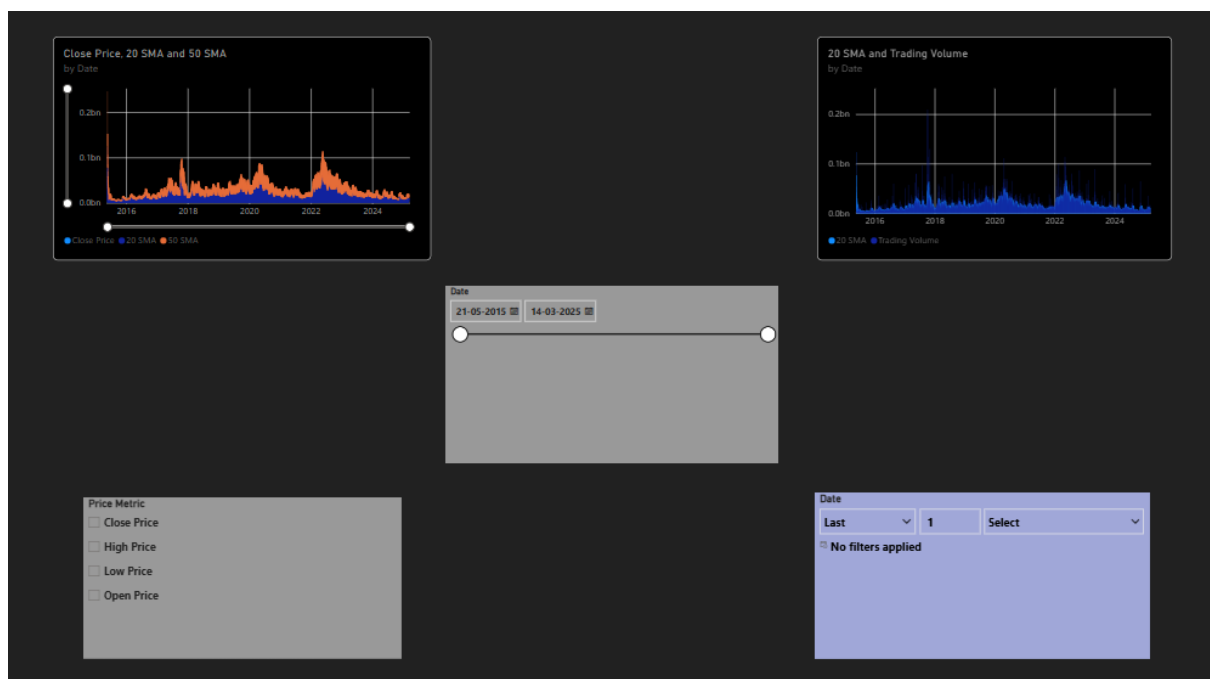
Low Price	Close Price	Trading Volume	Daily Return %
1.85	48.46	40912544200	0.00

9. Tooltip Settings

The Stock_Tooltip report page was connected to the main Close Price, 20 SMA, and 50 SMA chart. The tooltip type was set to Report page and Stock_Tooltip was selected as the tooltip page.



Final Dashboard



Open Price	High Price	Low Price	Close Price	Trading Volume
48.46	176.29	1.85	48.46	409125

